

“Care Guide”

An Inclusive Hospital Service

By Four Power

Zhaojun Liu
Haonan Pang
Haoyu Zhang
Yu Zhou





For final presentation

4/6/2026

Our Team





For final presentation

4/6/2026

Contents



01 Introduction

02 Feasibility Analysis

03 App Features

04 First Prototype

05 Peer Review

06 Final Prototype

07 Future Work

01 Introduction

Barrier 1:
Complex Interactions and
Information Overload
(especially relevant for **Elderly People**)

Barrier 2:
Sensory Exclusion in Queue
Management Systems
(People with **hearing impairments**,
people with **visual impairments**)



01 *Introduction*

In China, many hospitals have now introduced **digital services**, meaning that users often have to use the hospitals' online service platforms to avoid queuing in person or to book appointments with doctors.

However, we have found that older people face barriers to using many of these online service platforms due to factors such as **declining physical mobility**, **deteriorating eyesight** and **a lack of digital literacy**.

Overall, we aim to focus on the '**appointment booking**' and '**consultation**' stages to develop a **WeChat Mini Programme** for hospital digital platforms that is user-friendly for **older people**.

02 Feasibility Analysis

During the feasibility analysis phase, we primarily investigated the technology options available for mainstream hospital digital platforms and assessed their technical feasibility.

There are 2 main points here:

1. What is **WeChat Mini Programme** and why we choose it?

2. Can third-party digital healthcare platforms be seamlessly integrated into existing national healthcare information systems?

02 Feasibility Analysis

What is **WeChat Mini Programme** ?

Put simply, a WeChat Mini Programme is a **lightweight application** that can be used without the need to download or install it. It is embedded within the WeChat super-app, and users can open it directly by '**scanning a QR code**' or '**searching**' for it.

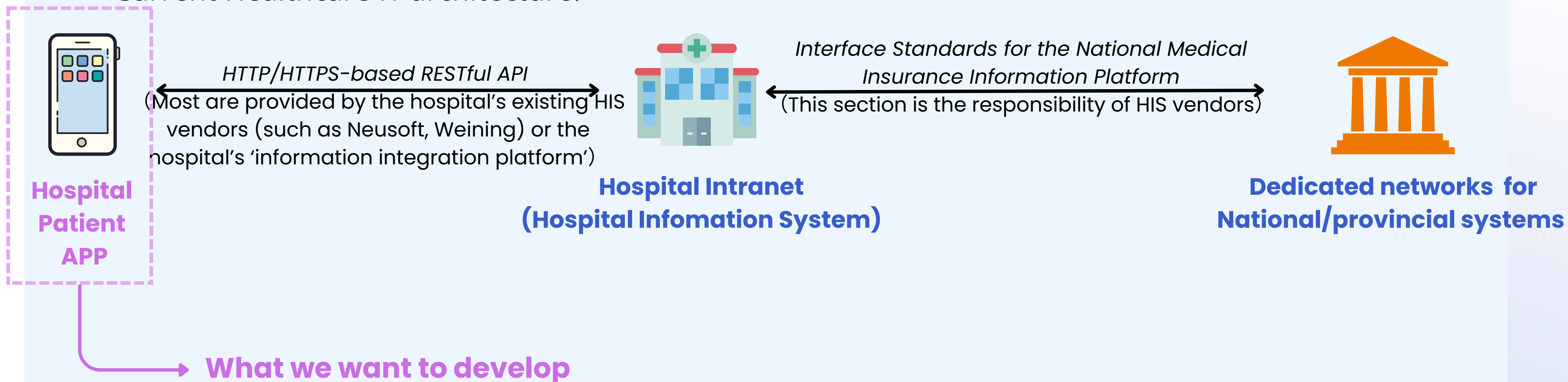
Why we choose it?

- Logging in to the App Store and Downloading is **a difficult task** for the elder people.
- Most Elder people **have Wechat** and know how to "Scan QR code".
- From the user's perspective, visiting a hospital is a **low-frequency activity**; having to download a new, rarely used app every time you visit a different hospital **makes for a very poor user experience**.

02 Feasibility Analysis

Can third-party digital healthcare platforms be seamlessly integrated into existing national healthcare information systems?

- Current Healthcare IT architecture:





For final presentation

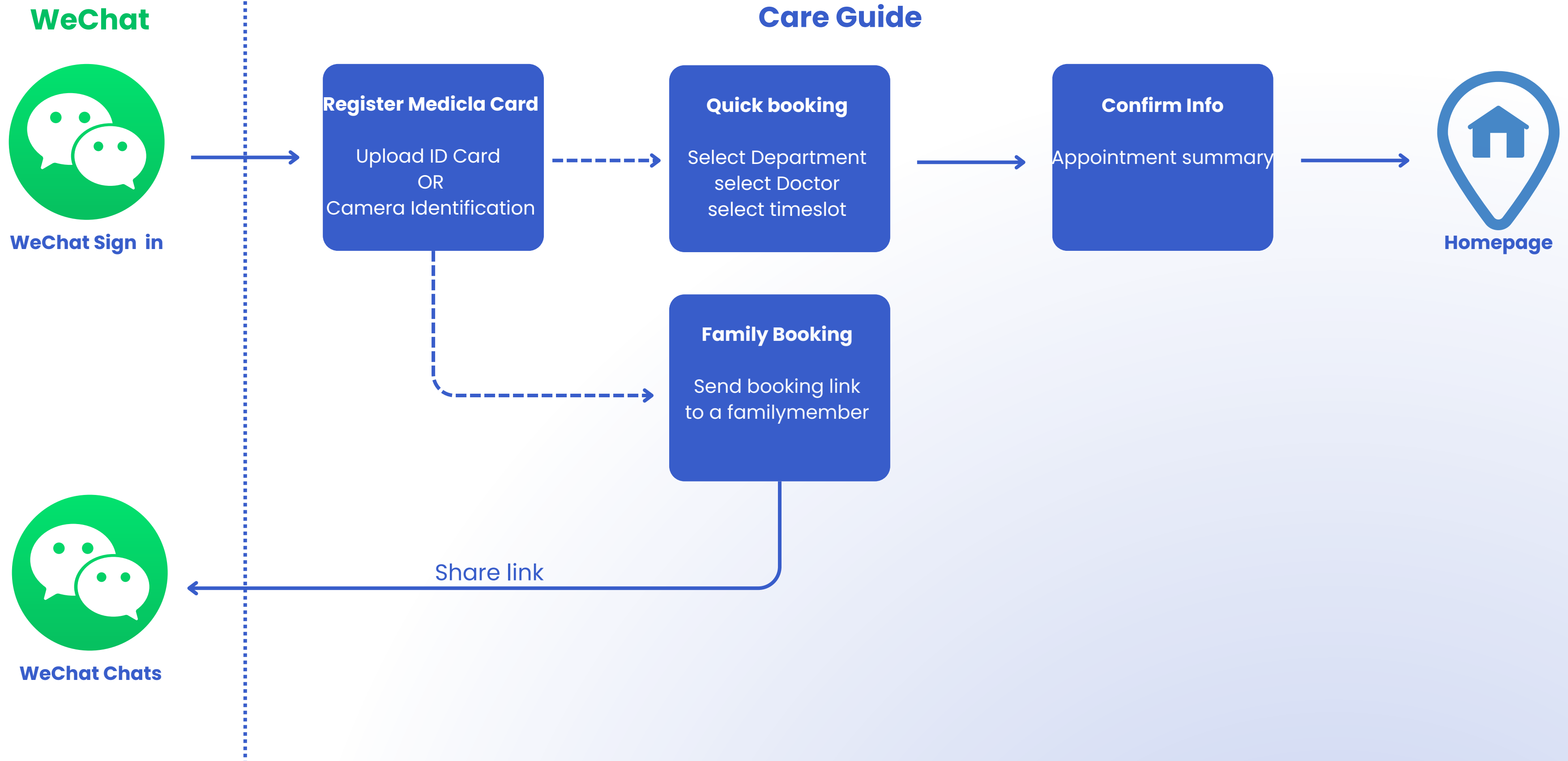
4/6/2026

03 Features

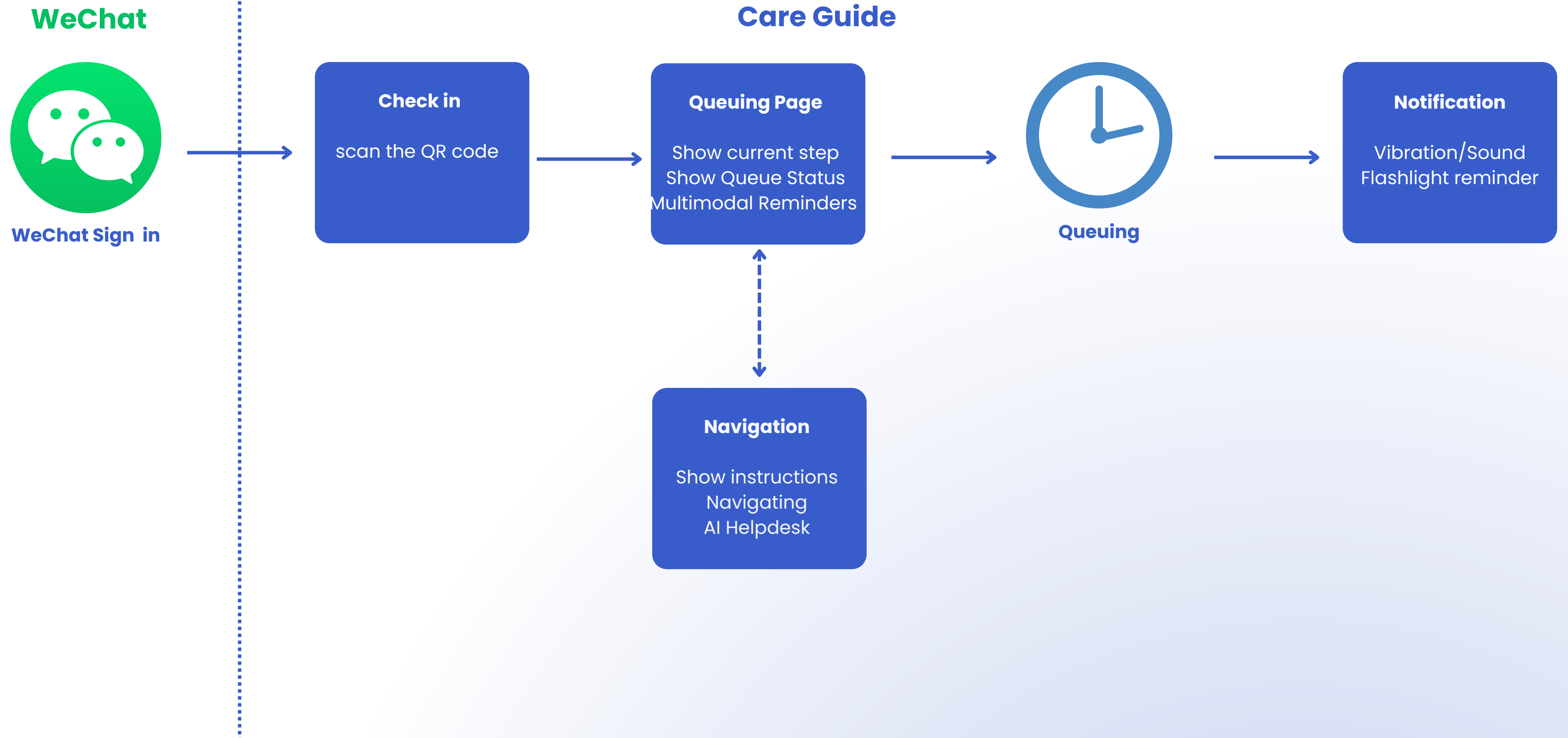
Two Main Functionalities:

- Appointment Booking
- Patient Guide

User Journey -- Appointment Booking



User Journey -- Patient Guide





03 Features

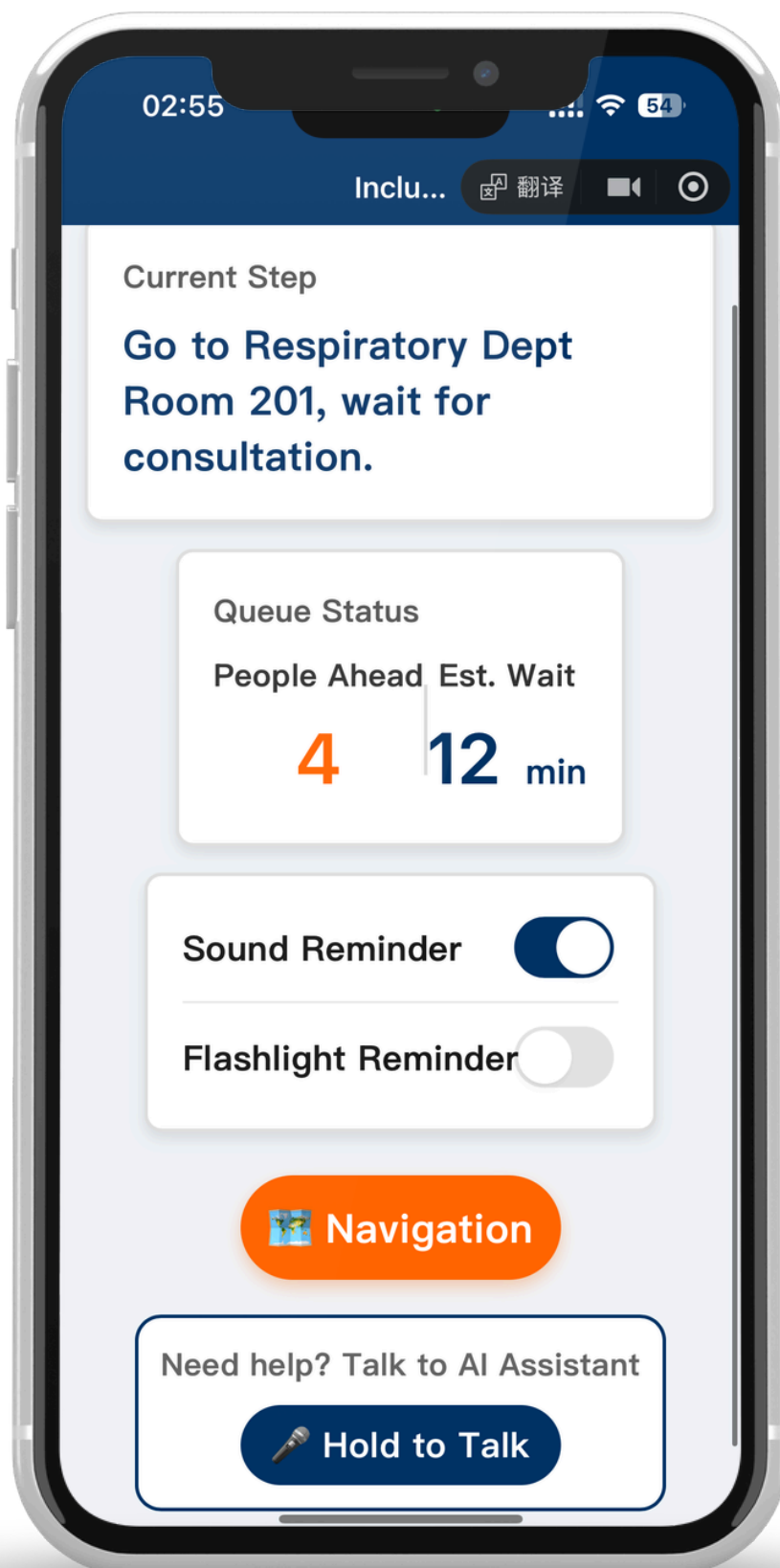
~~Besides Two Main Functionalities:~~

- ~~• Appointment Booking~~
- ~~• Patient Guide~~

Other Function point:

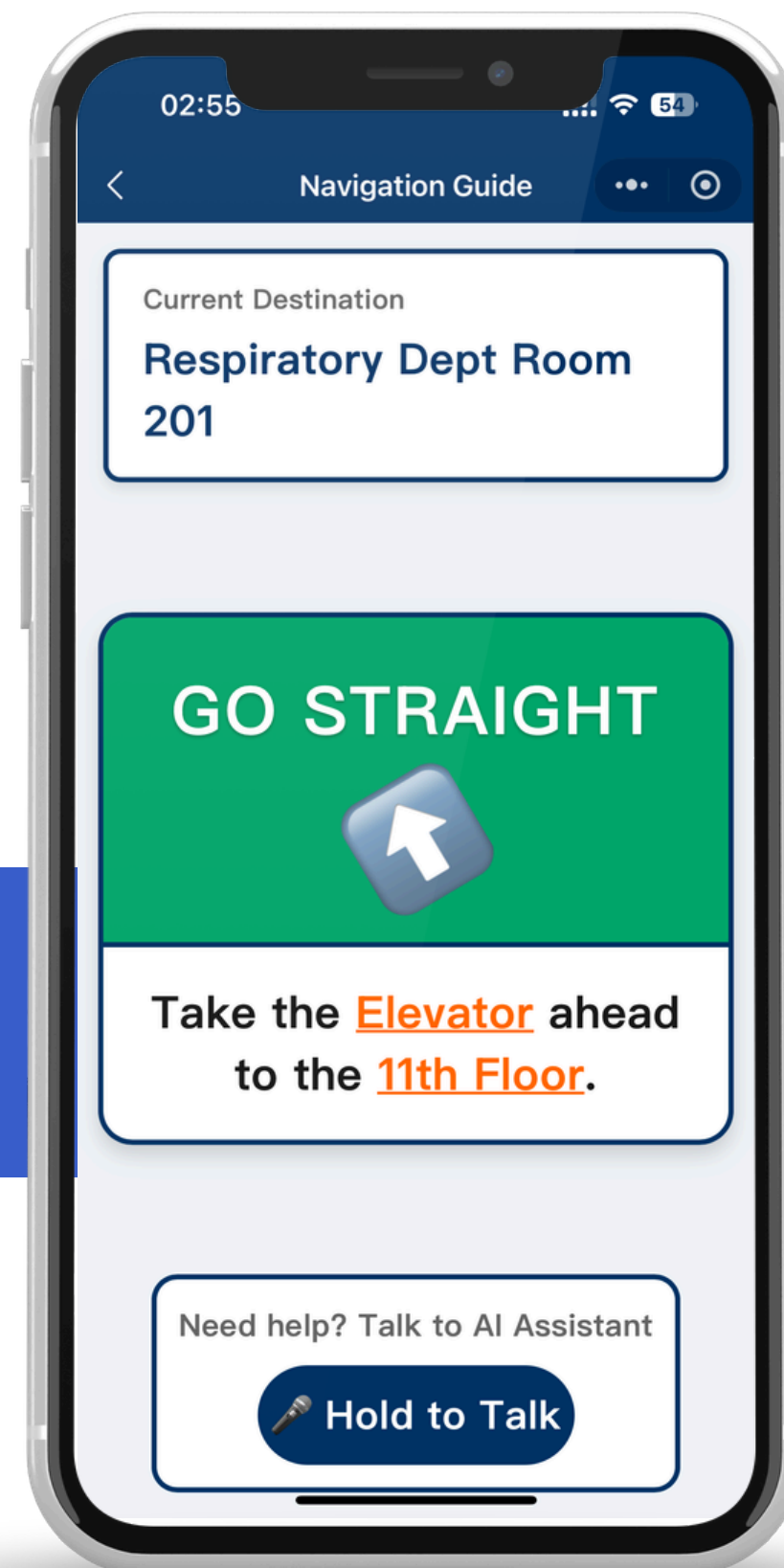
- **AI Assistant**

04 First Prototype - (very first)



Queueing Page

Navigation Page



04 Evolution Phase-Domain Expert

Ministry of Industry and Information Technology (MIIT). (2021). **“General Design Specification for Age-Friendly Mobile Internet Applications (Apps)”**. Beijing, China.

<http://wza.isc.org.cn/bztx/bzjd/mobile/index.html>

- Icons Accompanied by Text Labels (Combination of Graphics and Text)
- Removal of Advertisements
- Flat Information Architecture
- Large Typography and High Contrast Ratio
- Simplified Interaction Gestures
- . . .

04 Evolution Phase-Domain Expert

Development and validation of social compensation design scale for urban older users in the context of smart-home social media

DIGITAL HEALTH
Volume 12: 1–27
© The Author(s) 2026
Article reuse guidelines:
sagepub.com/journals-permissions
DOI: 10.1177/20552076251411270
journals.sagepub.com/home/dhj


Ke Ma^{1,2} , Ying Zhao³ , Francesco Ermanno Guida⁴ , Meng Gao⁵ , Renke He⁵
and Jinjun Xia^{1,2} 

Table 12. Social Compensation Design Scale (SCDS).

Constructs	Measurement items
UIQ	UIQ-1. I feel that the text features, icon features, image features, and colors of the smart-home social media interfaces meet my needs during my use
	UIQ-2. I pay attention to the text features, icon features, image features, and colors of smart-home social media interfaces
	UIQ-5. I think the rationality of the functional layout, hierarchical structure, and spacing of the smart-home social media interfaces affects my usage experience (such as being happier)
IQ	IQ-7. I find it natural and easy to wake up to smart-home social media, and it responds to me on time
	IQ-8. I think the voice interaction of smart-home social media can meet my needs, and it is natural and responsive
	IQ-10. I think the operation of smart-home social media is very simple, and it is not easy to make mistakes
CQ	CQ-12. I think using smart-home social media can help me better integrate into the group
	CQ-17. I feel that smart-home social media can help me share information with others, which is what I need
	CQ-19. I think using smart-home social media can help find more old friends
	CQ-20. I feel that by using smart-home social media, I can track my friends' activities in real time.
	CQ-21. I feel that when using smart-home social media, we share a common goal and passion for helping others solve their problems
SQ	CQ-23. I think it is convenient to use smart-home social media to generate sharable content and share it with friends
	SQ-26. I feel accompanied and cared for in using smart-home social media
	SQ-27. I think I can easily gain emotional satisfaction when using smart-home social media
	SQ-28. I feel that smart-home social media gives me a sense of belonging
	SQ-29. I feel that my life cannot do without smart-home social media

CQ: content quality; IQ: interaction quality; SQ: service quality; UIQ: user interface quality.



For final presentation

4/6/2026

04 Evolution Phase-Domain Expert

Development and validation of social compensation design scale for urban older users in the context of smart-home social media

Ke Ma^{1,2} , Ying Zhao³ , Francesco Ermanno Guida⁴ , Meng Gao⁵ , Renke He⁵ and Jinjun Xia^{1,2} 

DIGITAL HEALTH
Volume 12: 1–27
© The Author(s) 2026
Article reuse guidelines:
sagepub.com/journals-permissions
DOI: 10.1177/20552076251411270
journals.sagepub.com/home/dhj


Some Insights:

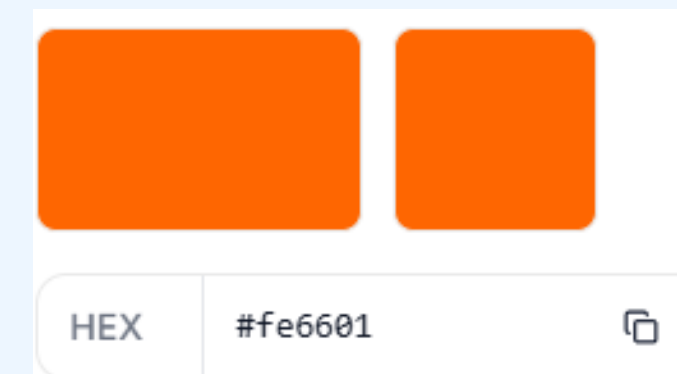
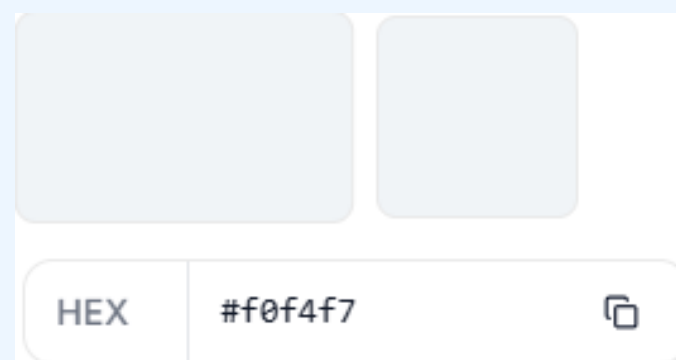
- We usually assume that large font sizes and high contrast in interfaces are intended to ensure ‘readability’, but assessment tools (UIQ-3, UIQ-5) emphasise that the appropriateness of visual elements (text, icons, colours) and typographic layout directly influences older people’s mood and sense of enjoyment.
- The IQ-10 scale emphasises that the operation is very simple and ‘error-proof’, whilst the IQ-8 scale highlights the natural and responsive nature of the voice interaction.

04 Evolution Phase

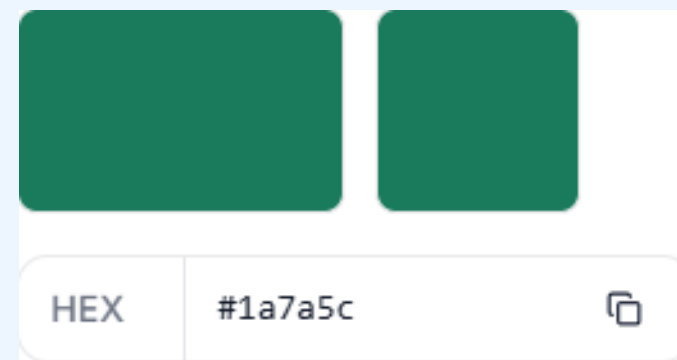
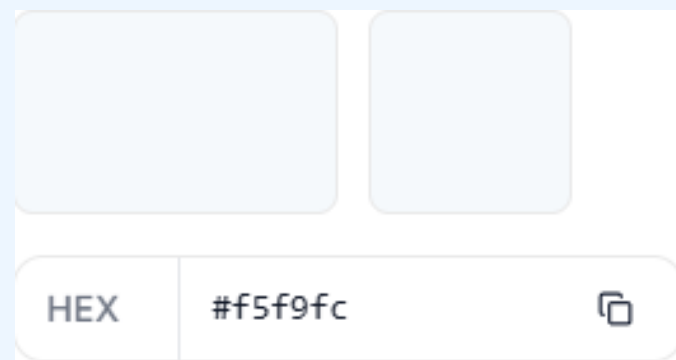
Compared with very first version:

- A **complete front-end** of all the expected features
- **Theme color changes:** (To ensure color consistency)

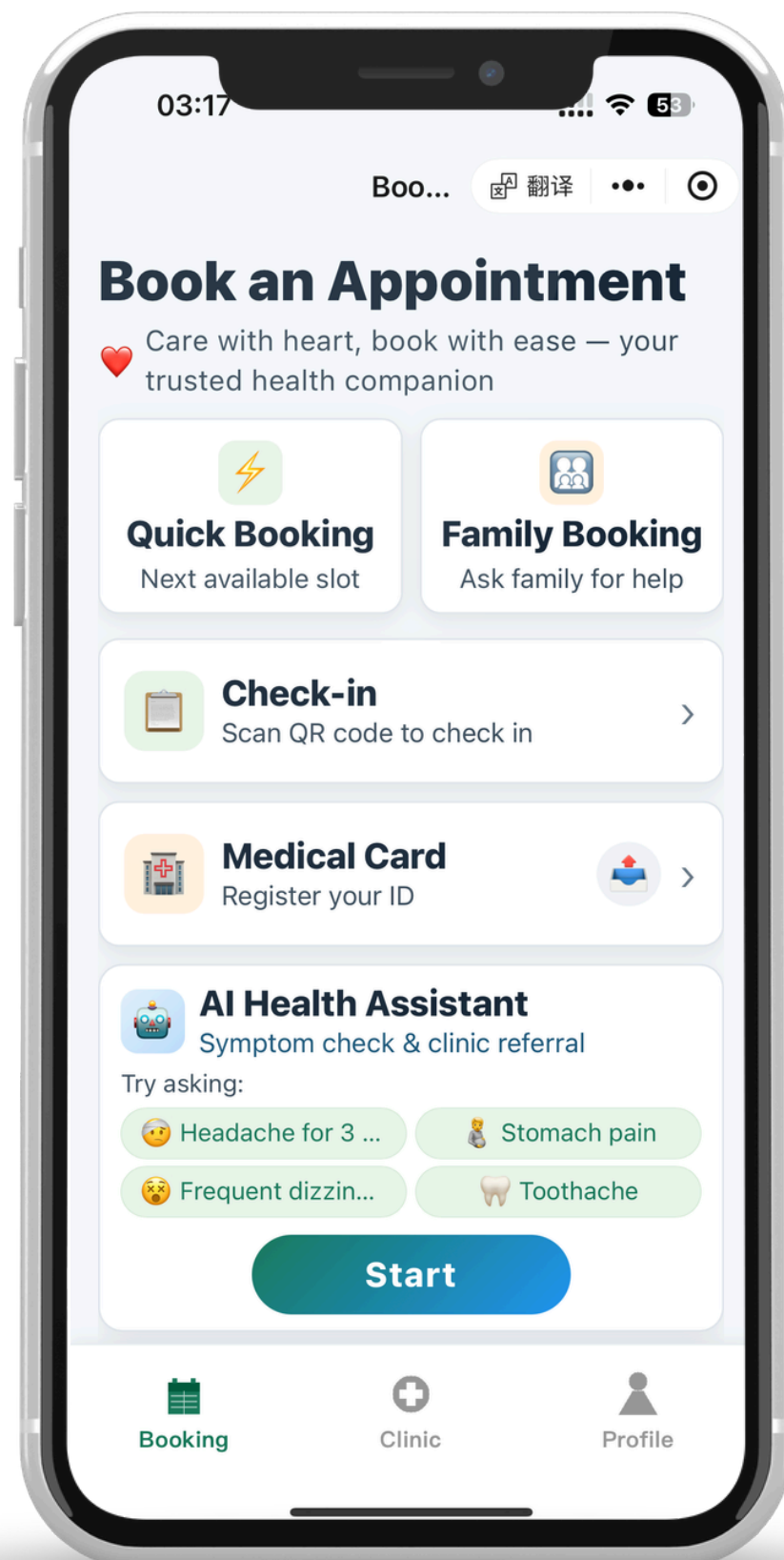
Previous:



Afterwards:

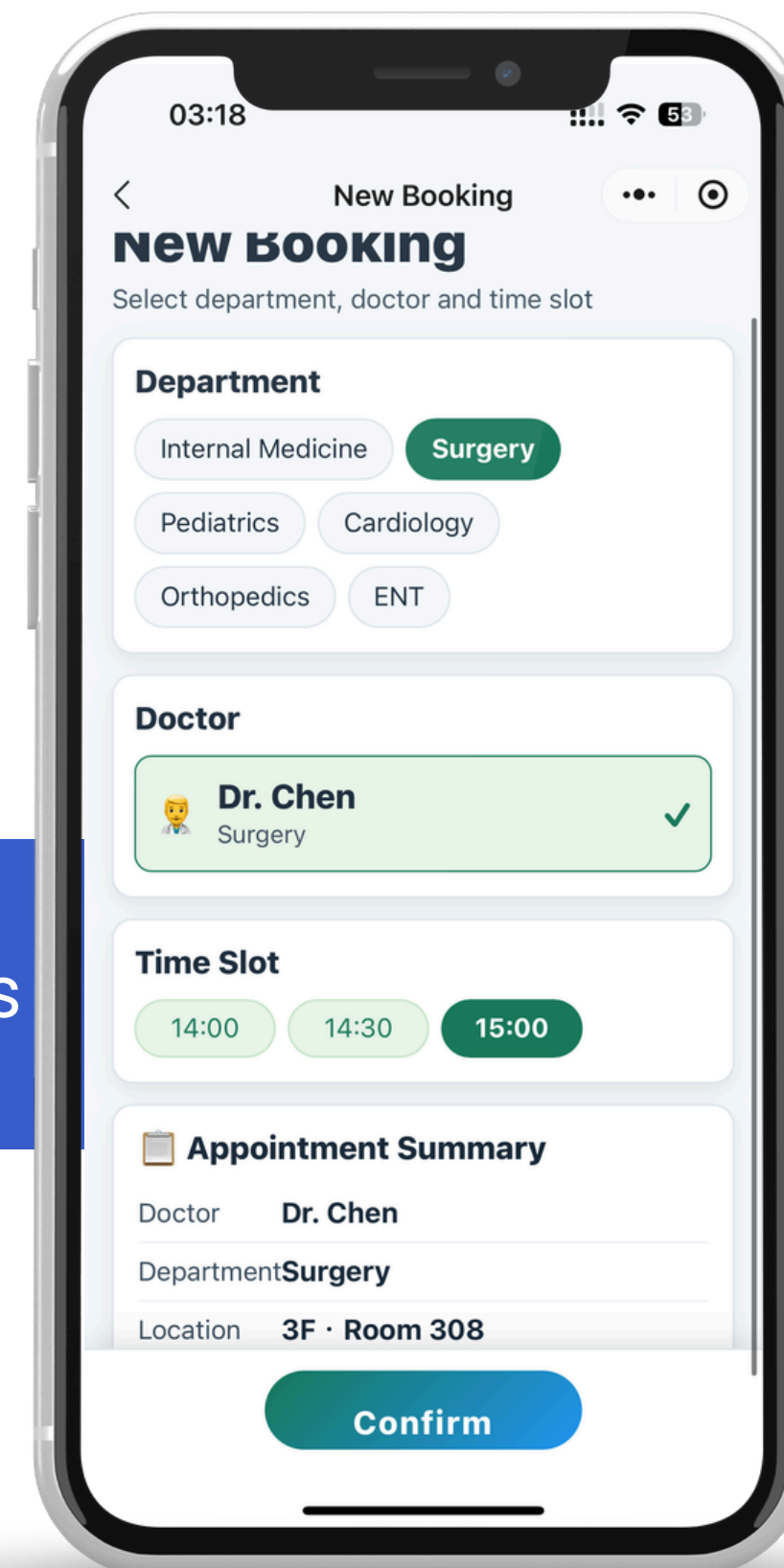


04 First Prototype - (Full version)



Home Page

Booking Details

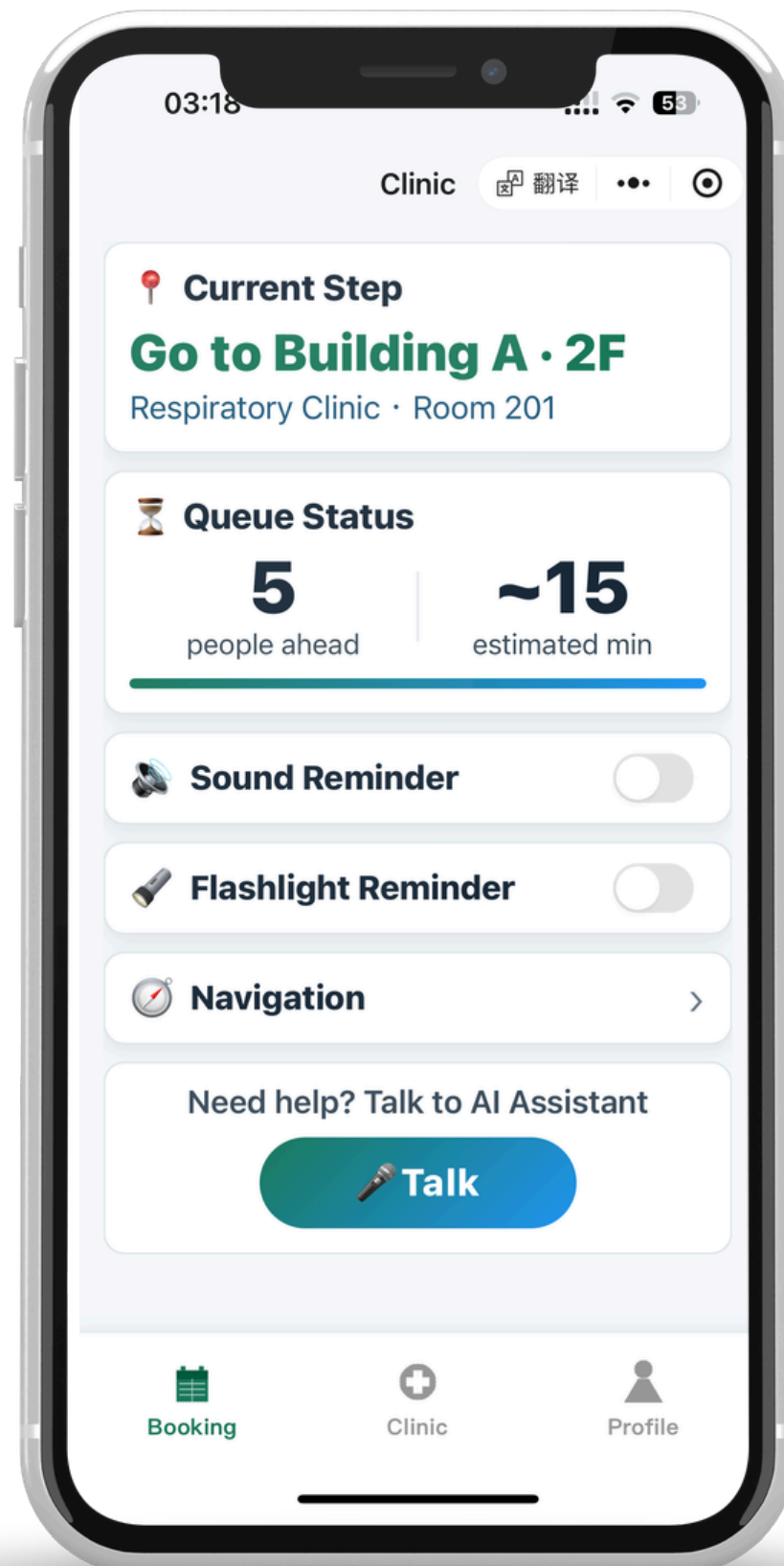




For final presentation

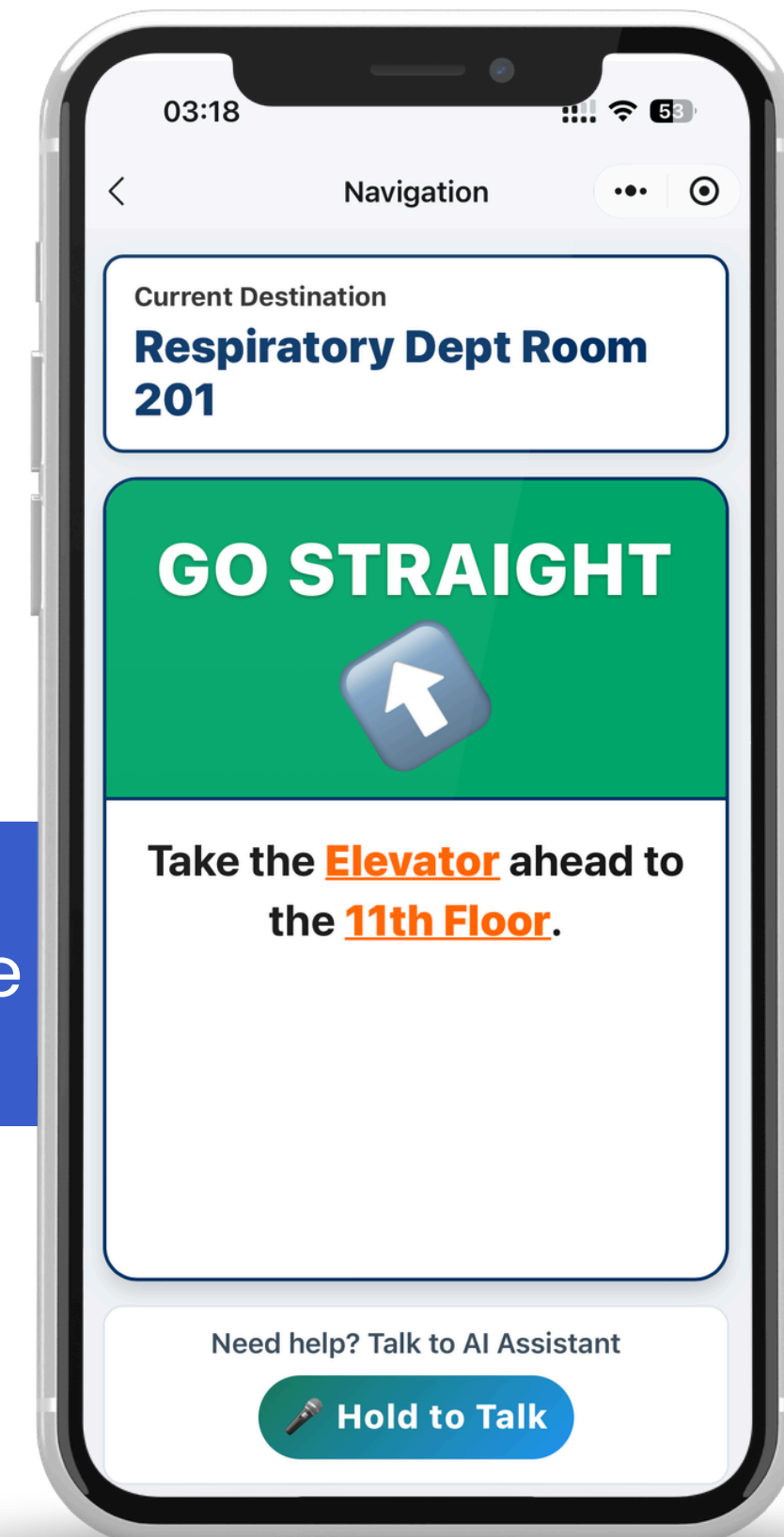
4/6/2026

04 First Prototype - (Full version)

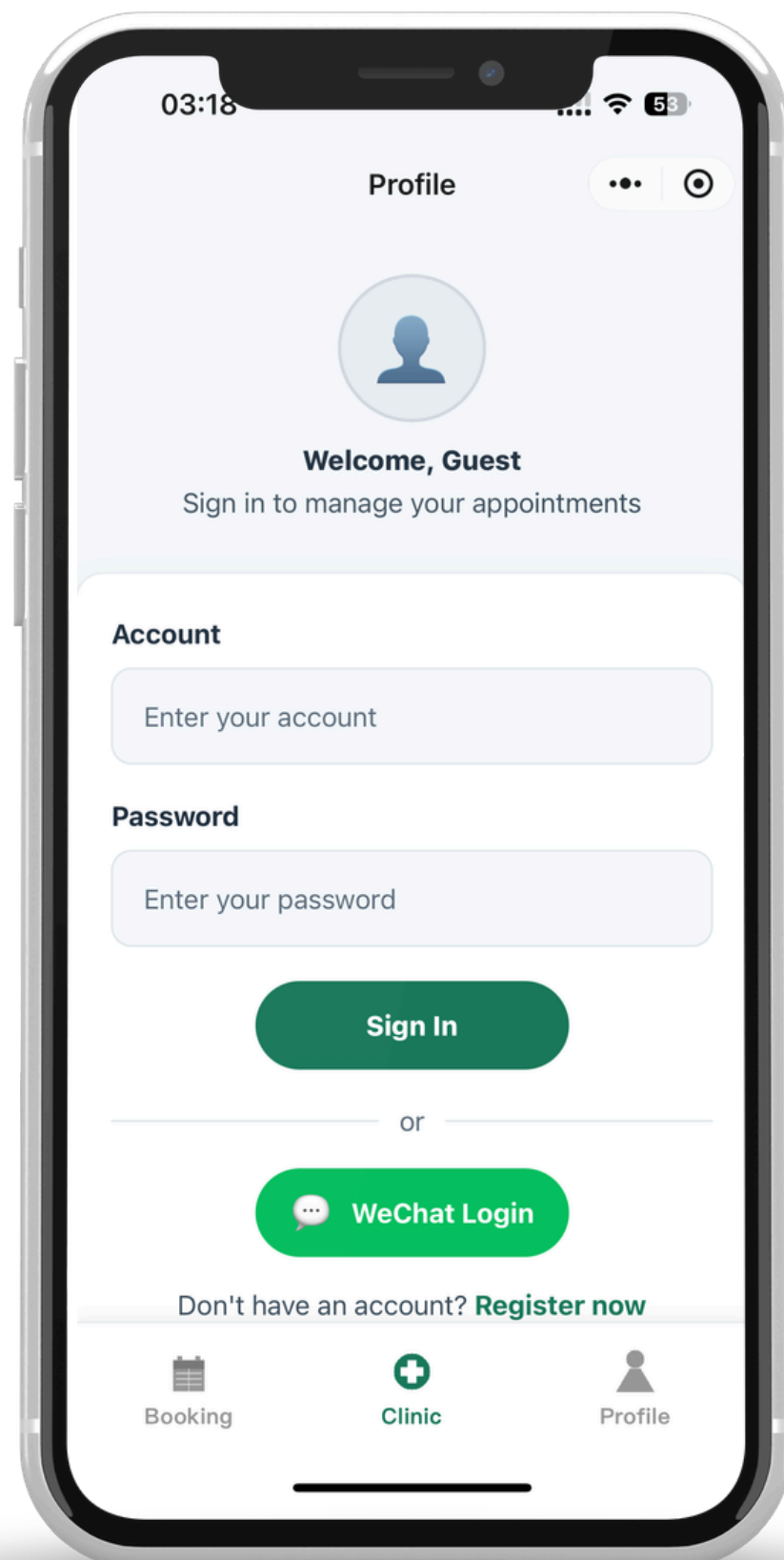


Queuing Status

Navigation Page

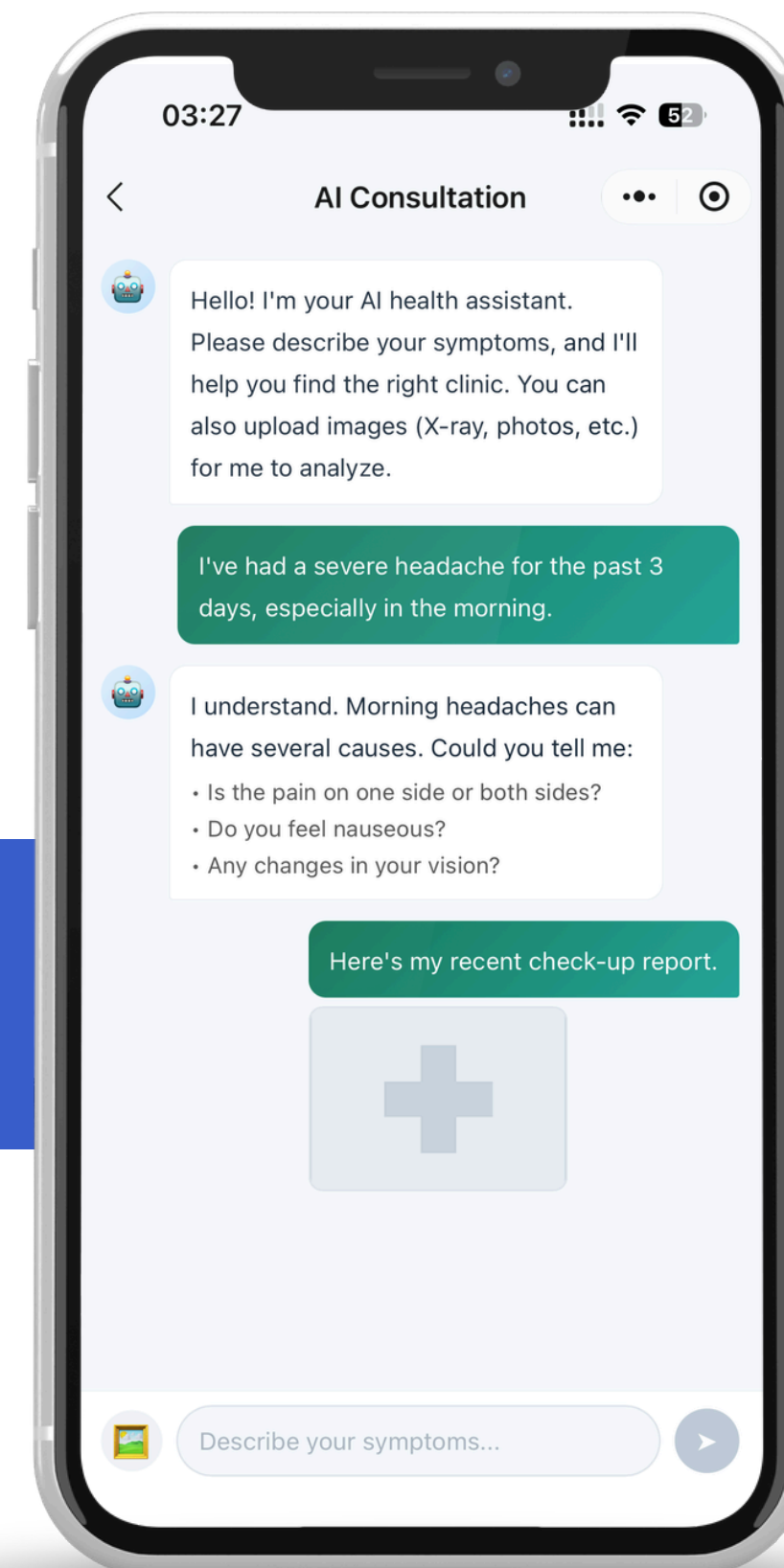


04 First Prototype - (Full version)



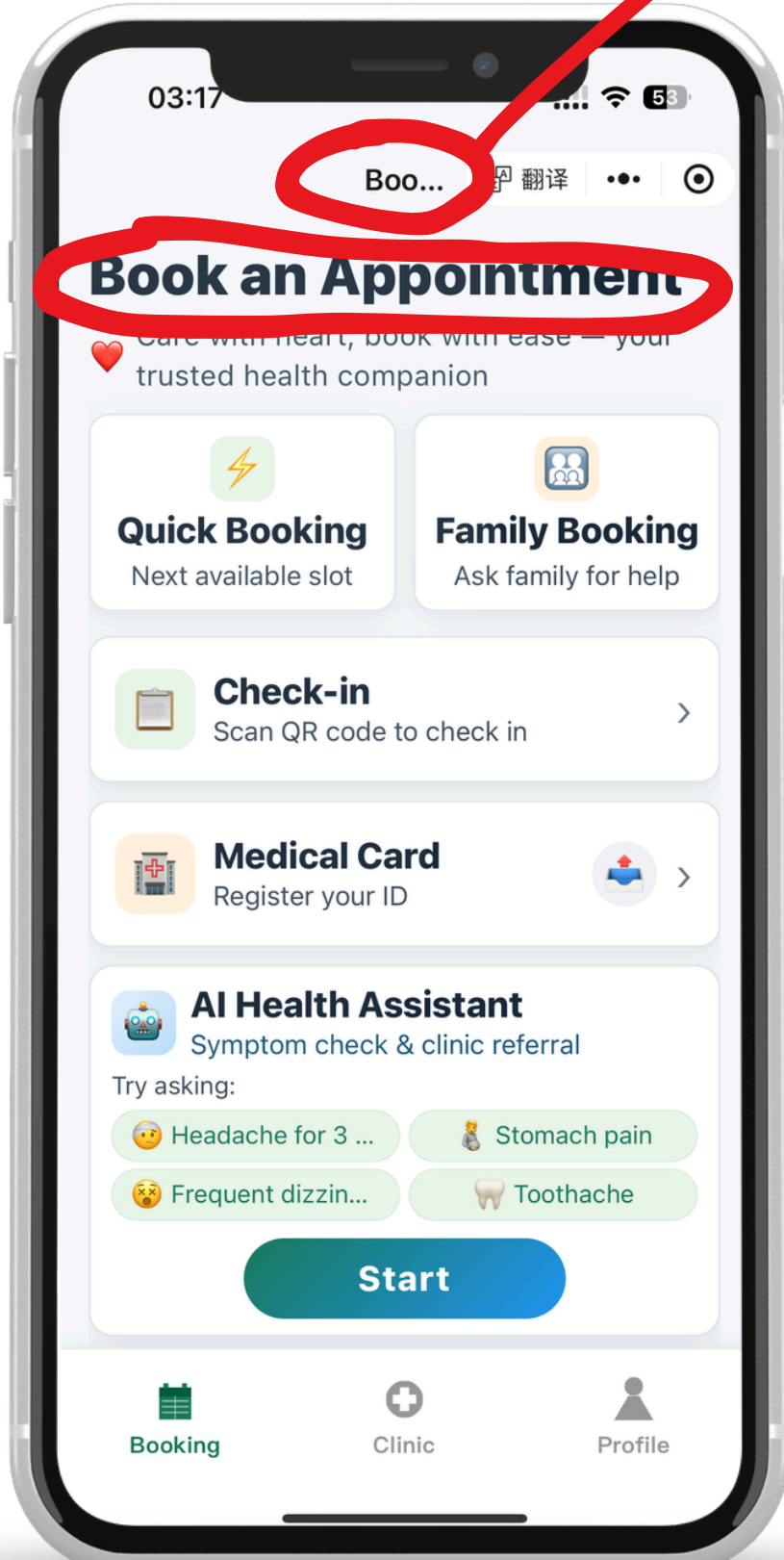
Profile Page

AI Assistant



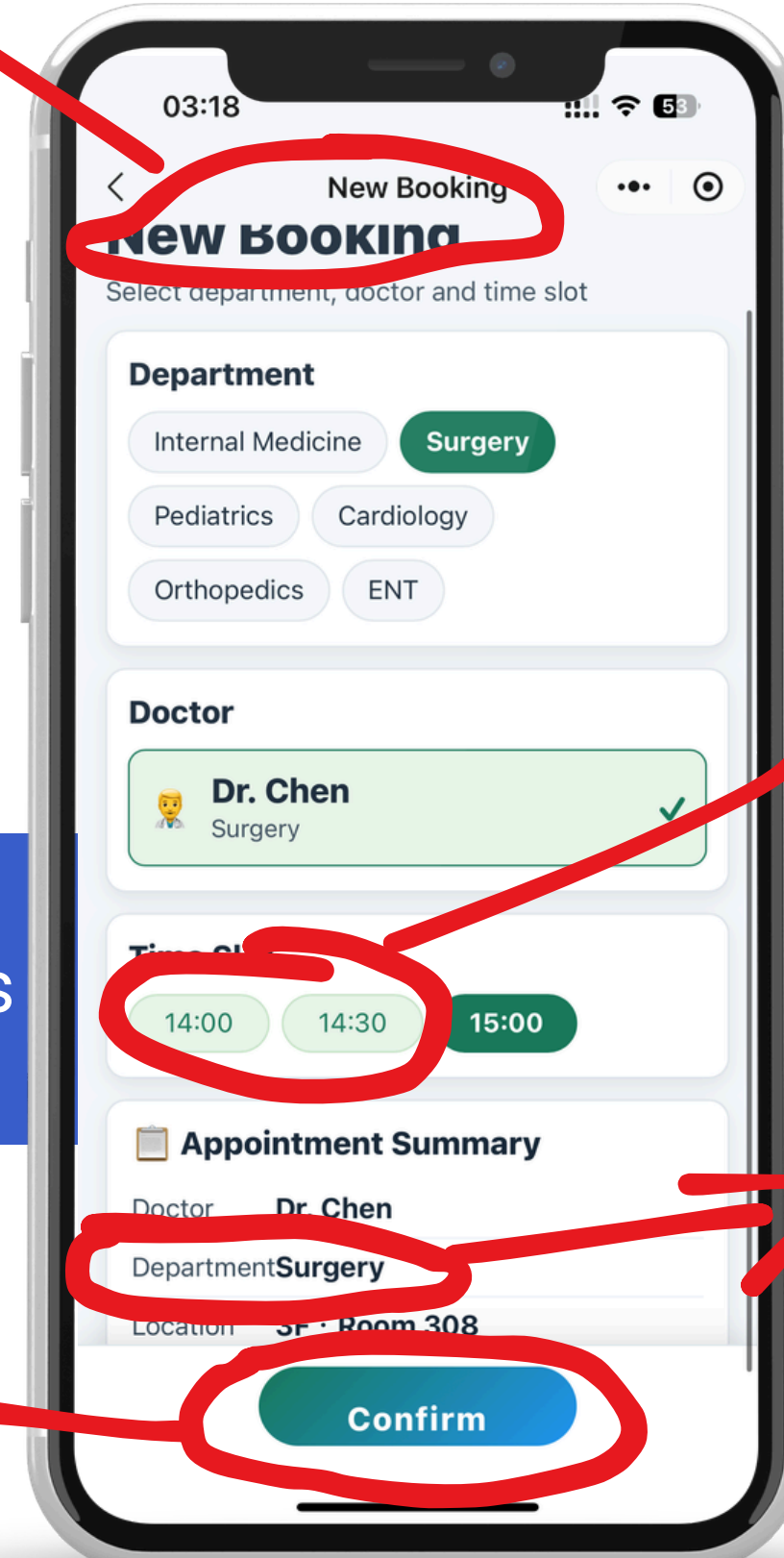
05 Peer Review

1.duplicated Title



Home Page

Booking Details



2.Low Contrast

3.Overlapping characters

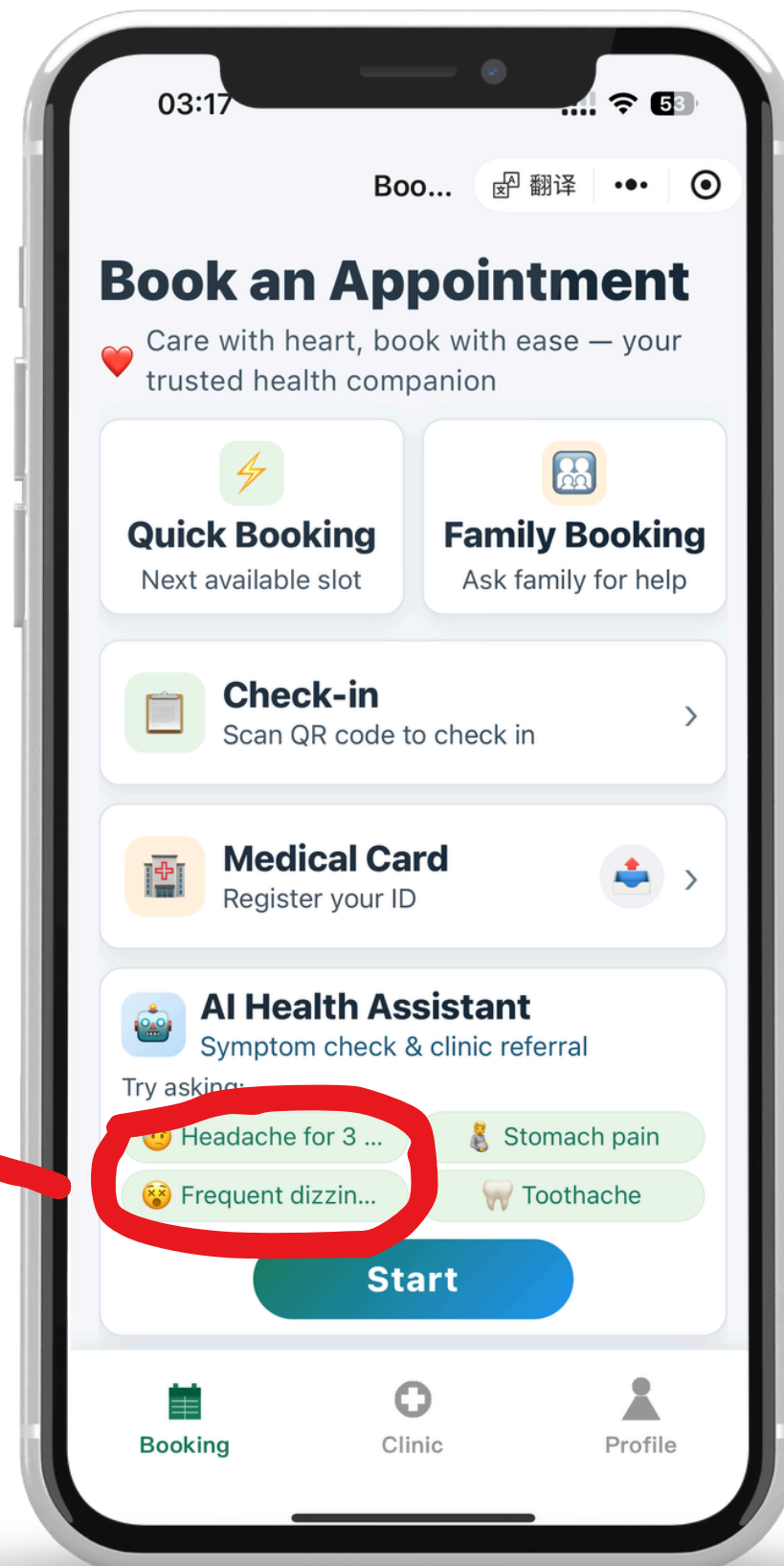
4.Misalignment



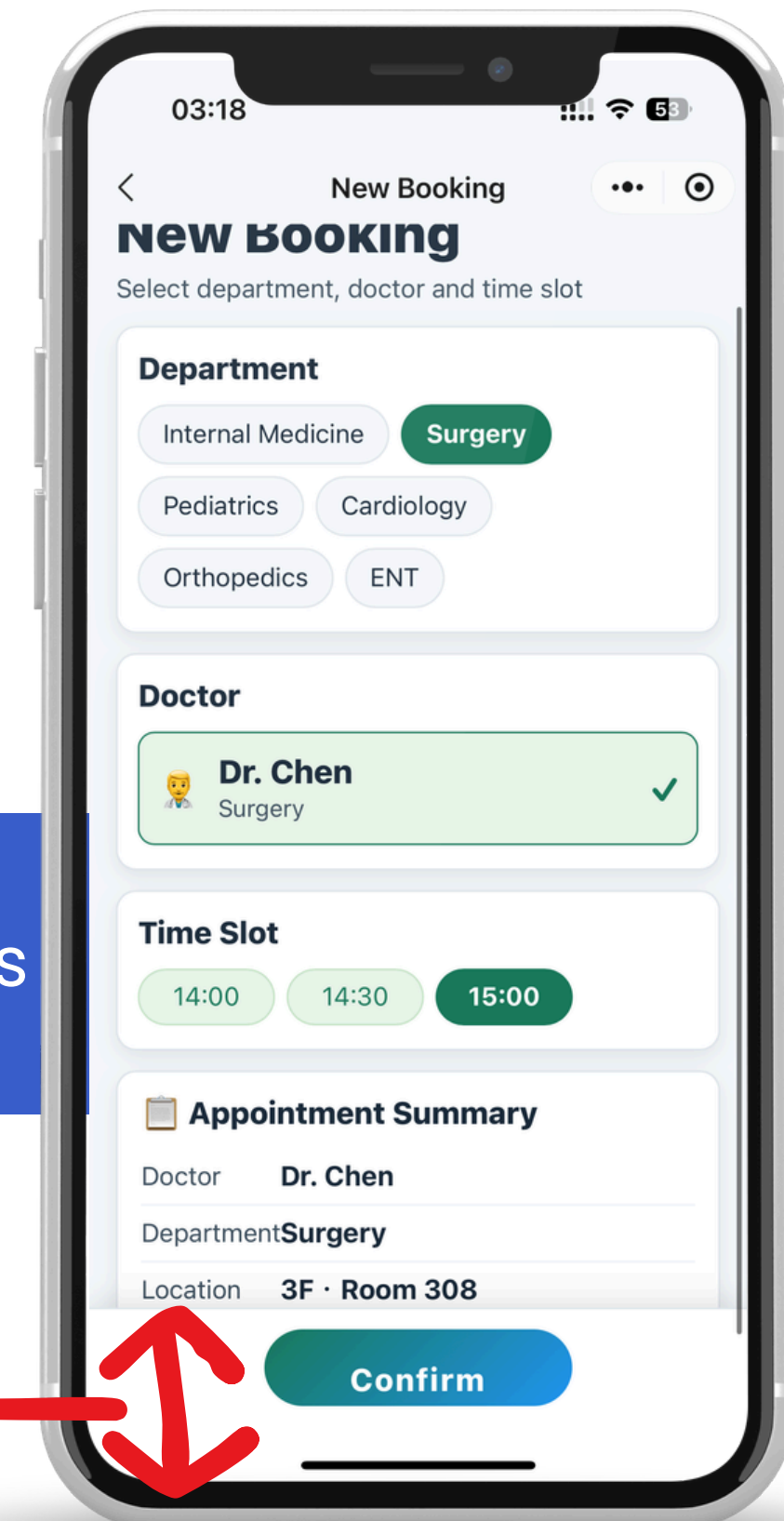
For final presentation

4/6/2026

05 Peer Review



Home Page



Booking Details

5. The text is truncated by '...'

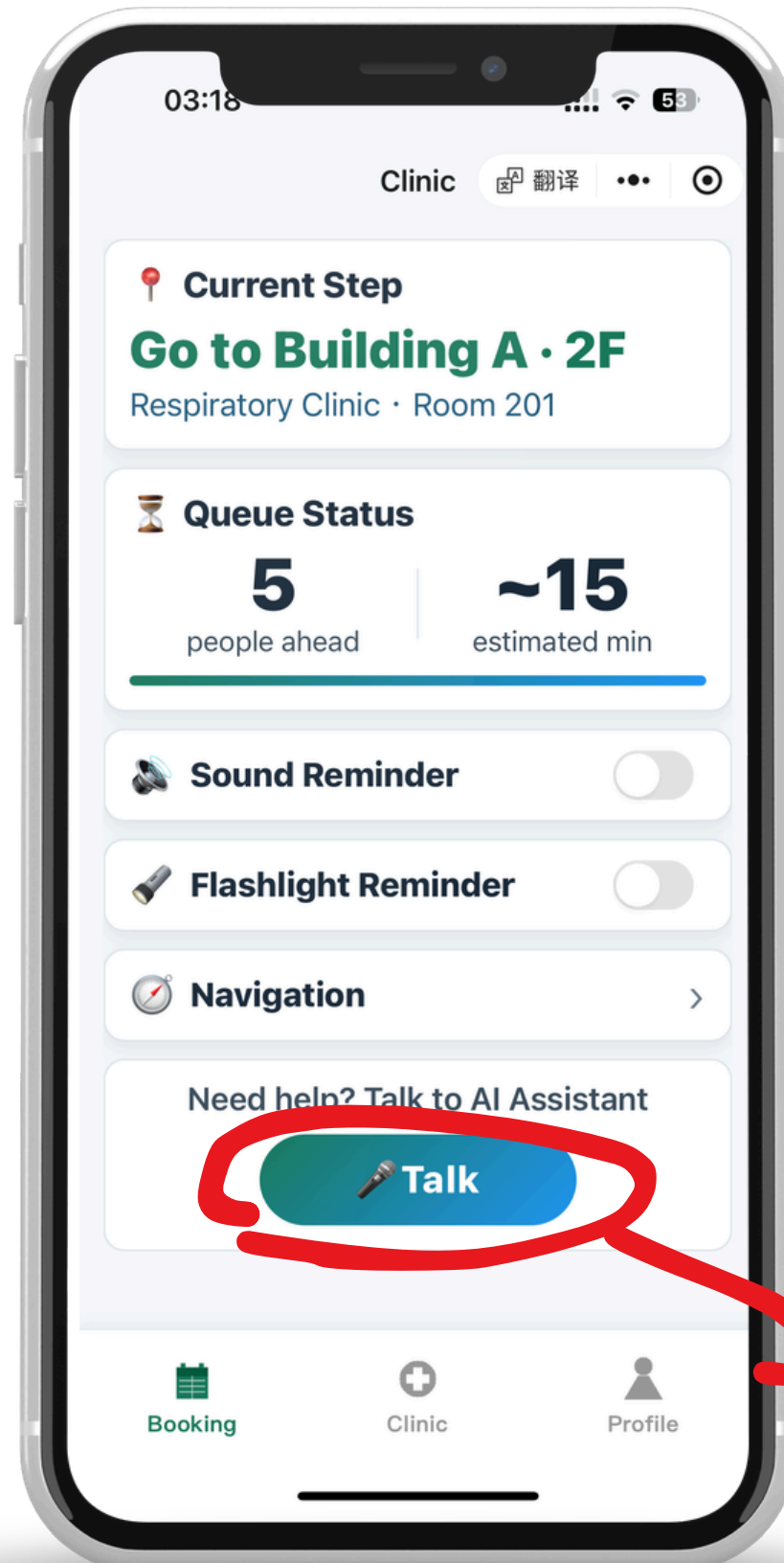
6. Too much white space



For final presentation

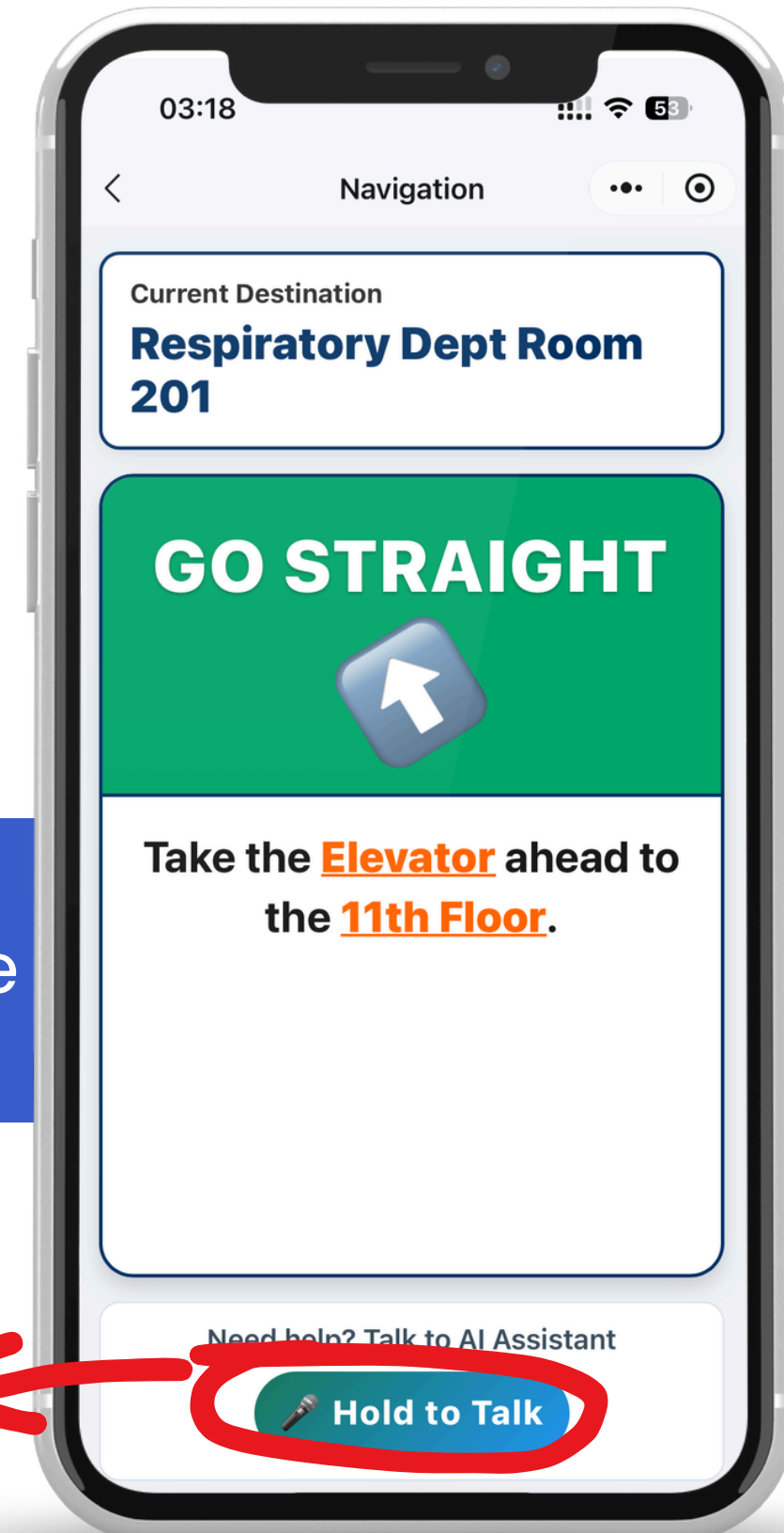
4/6/2026

05 Peer Review



Queuing Status

Navigation Page



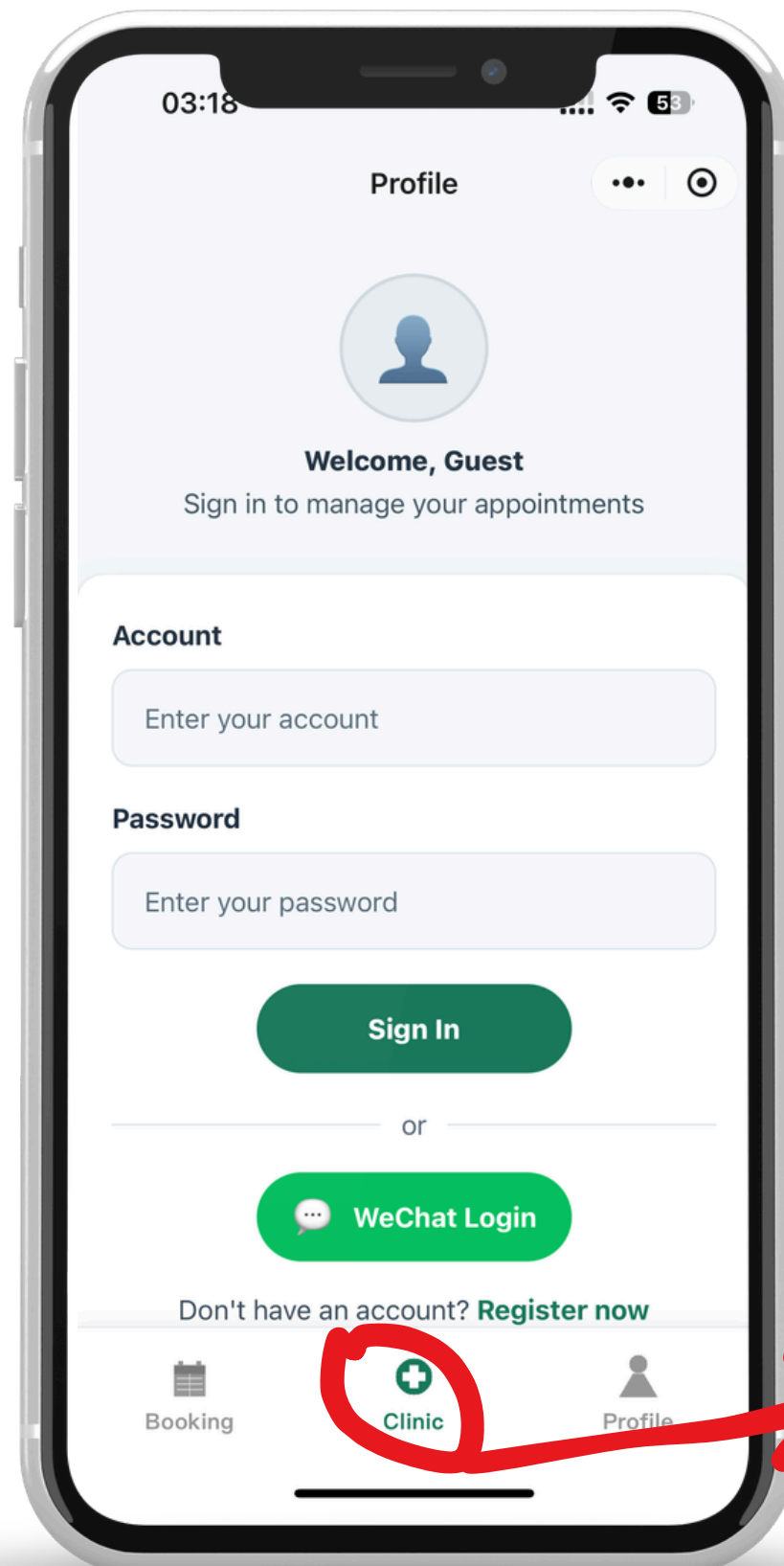
7. Inconsistent text on identical buttons



For final presentation

4/6/2026

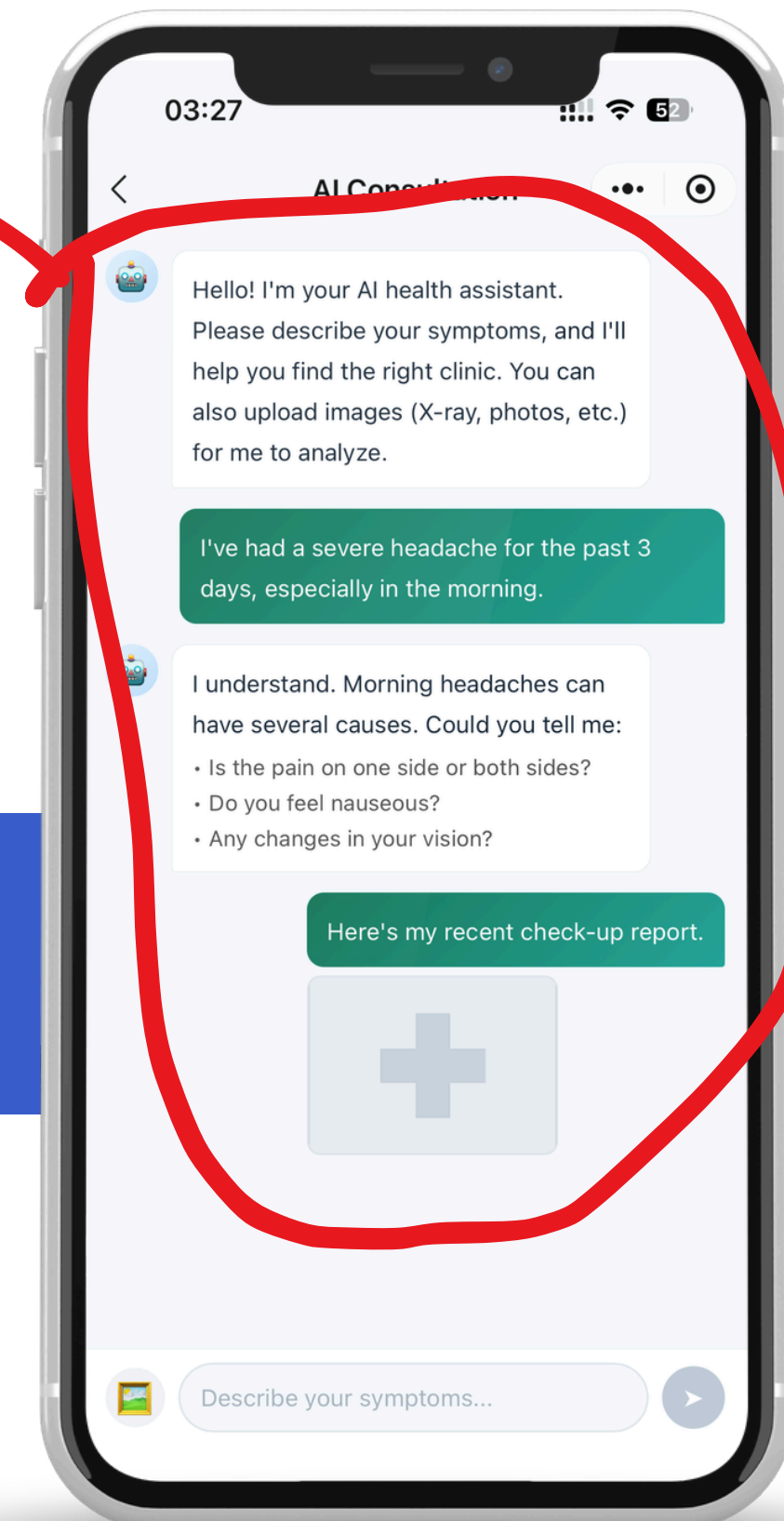
05 Peer Review



8. Font is too small

Profile Page

AI Assistant



9. The navigation bar color does not match the page

05 *Peer Review*

To sum up:

1.duplicated Title

2.Low Contrast

3.Overlapping characters

4.Misalignment

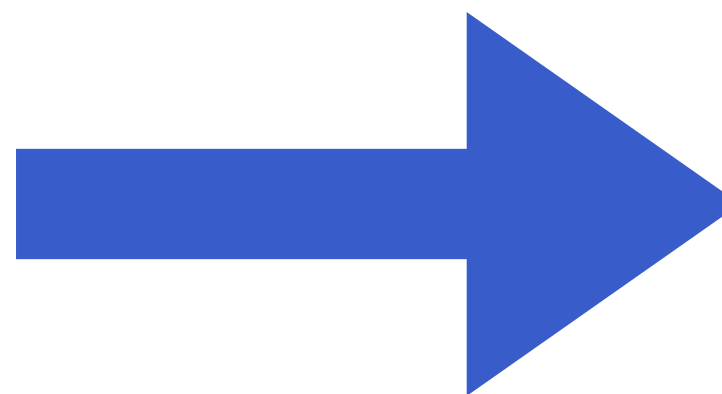
5.The text is truncated by '...'

6.Too much white space

7.Inconsistent text on identical buttons

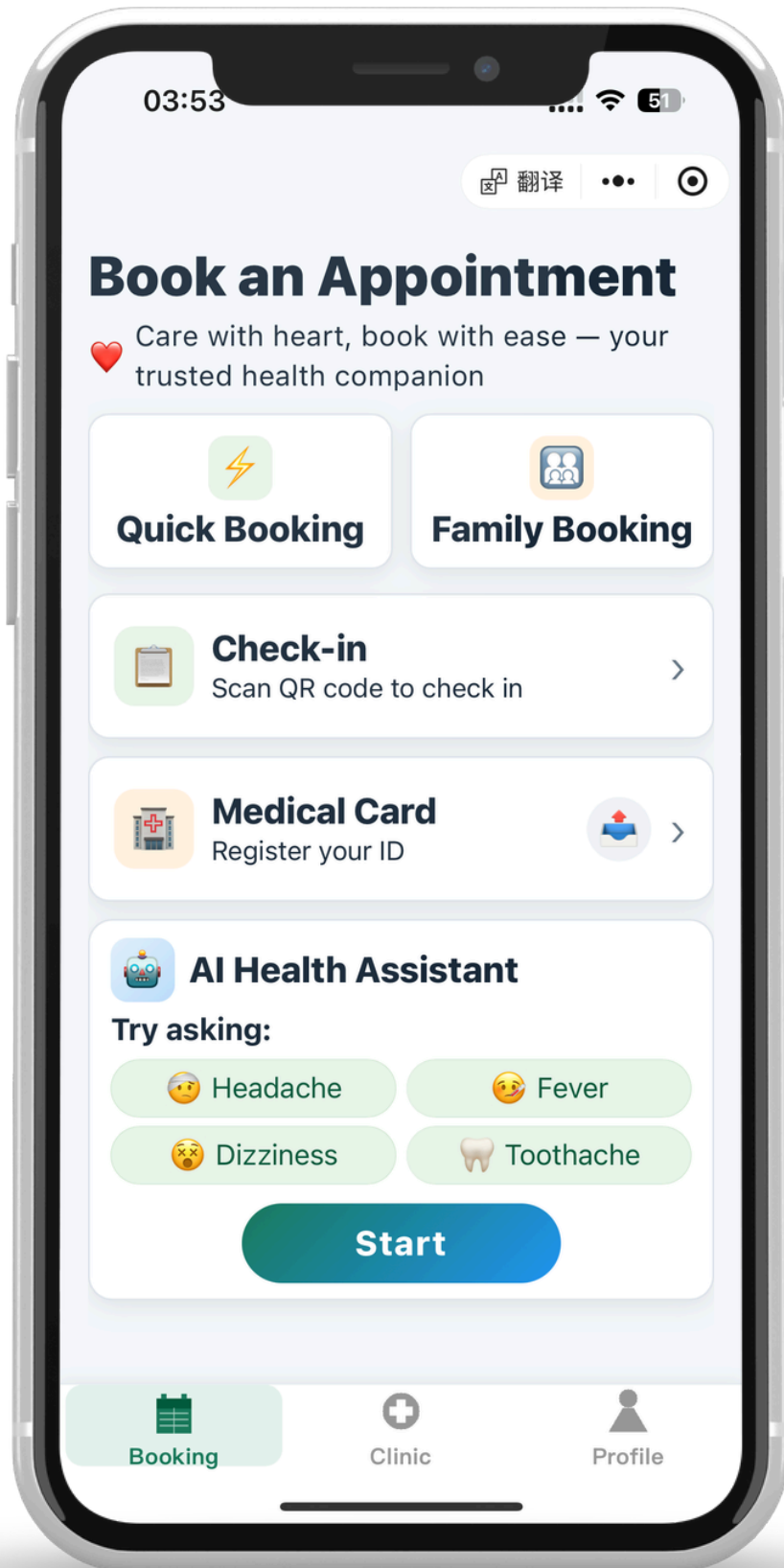
8.Font is too small

9.The navigation bar color does not match the page



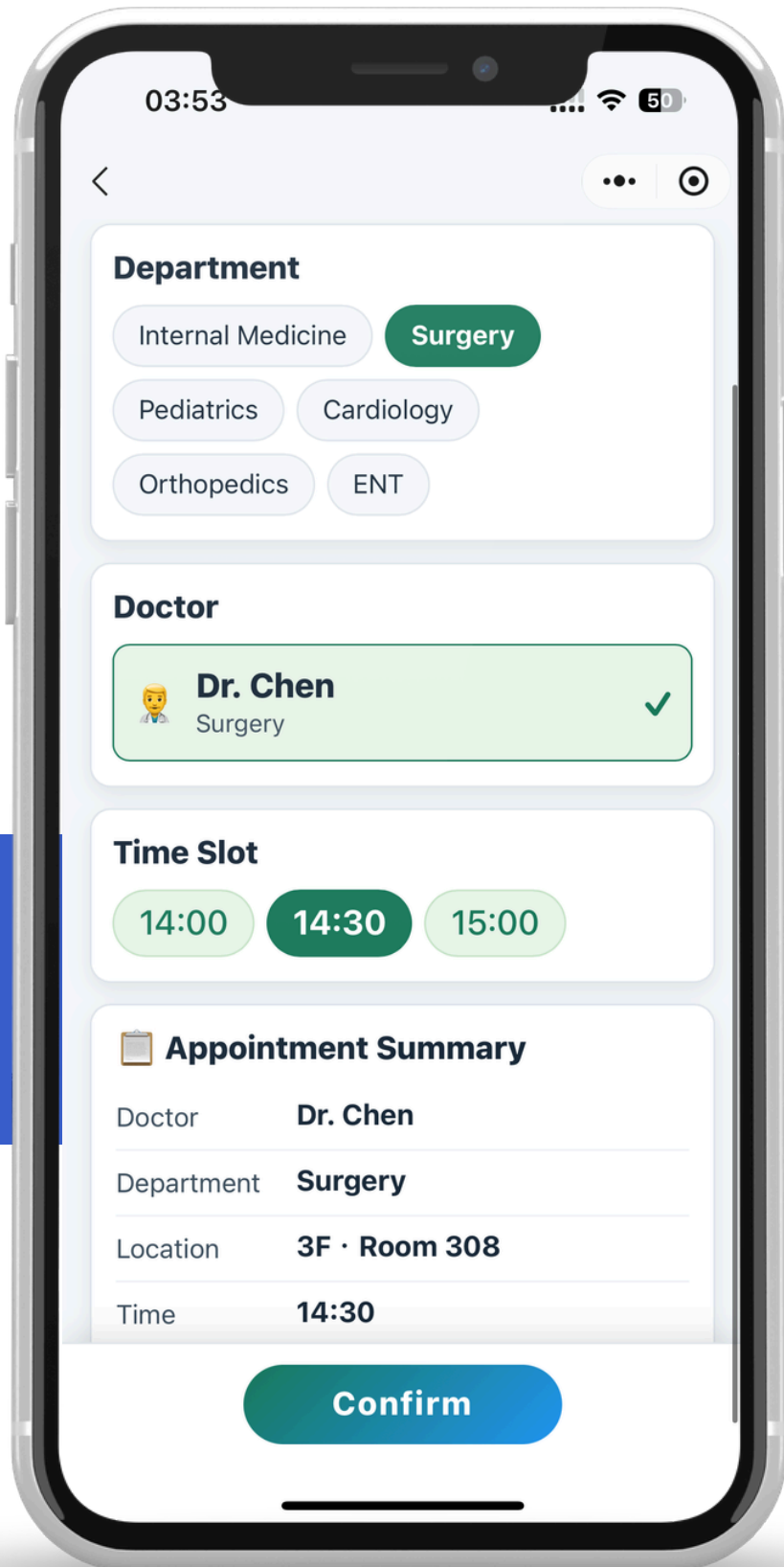
All issues have been resolved
in the **final version**

06 Final Prototype

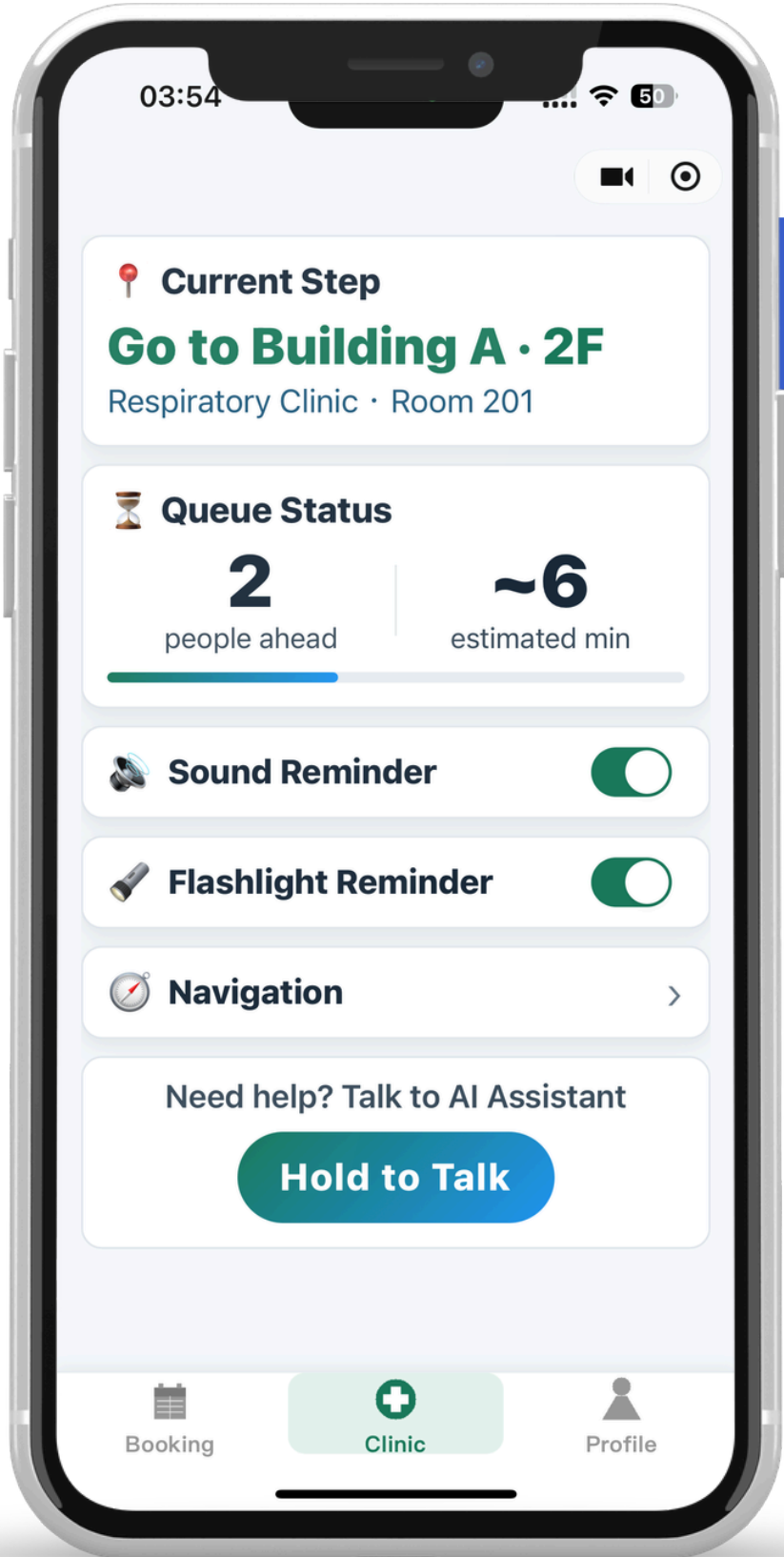


Home Page

Booking Details



06 Final Prototype

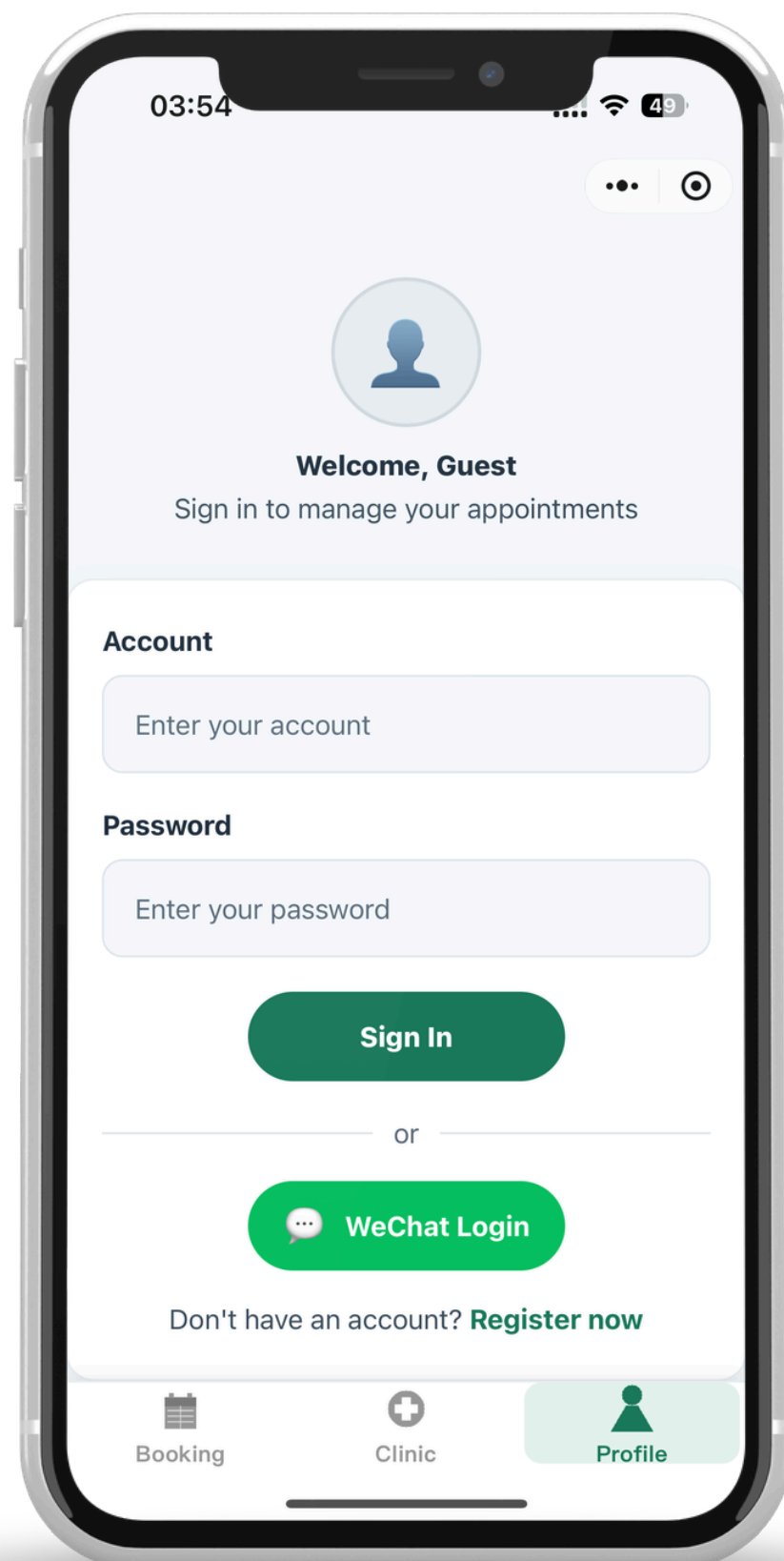


Queuing Status

Navigation Page

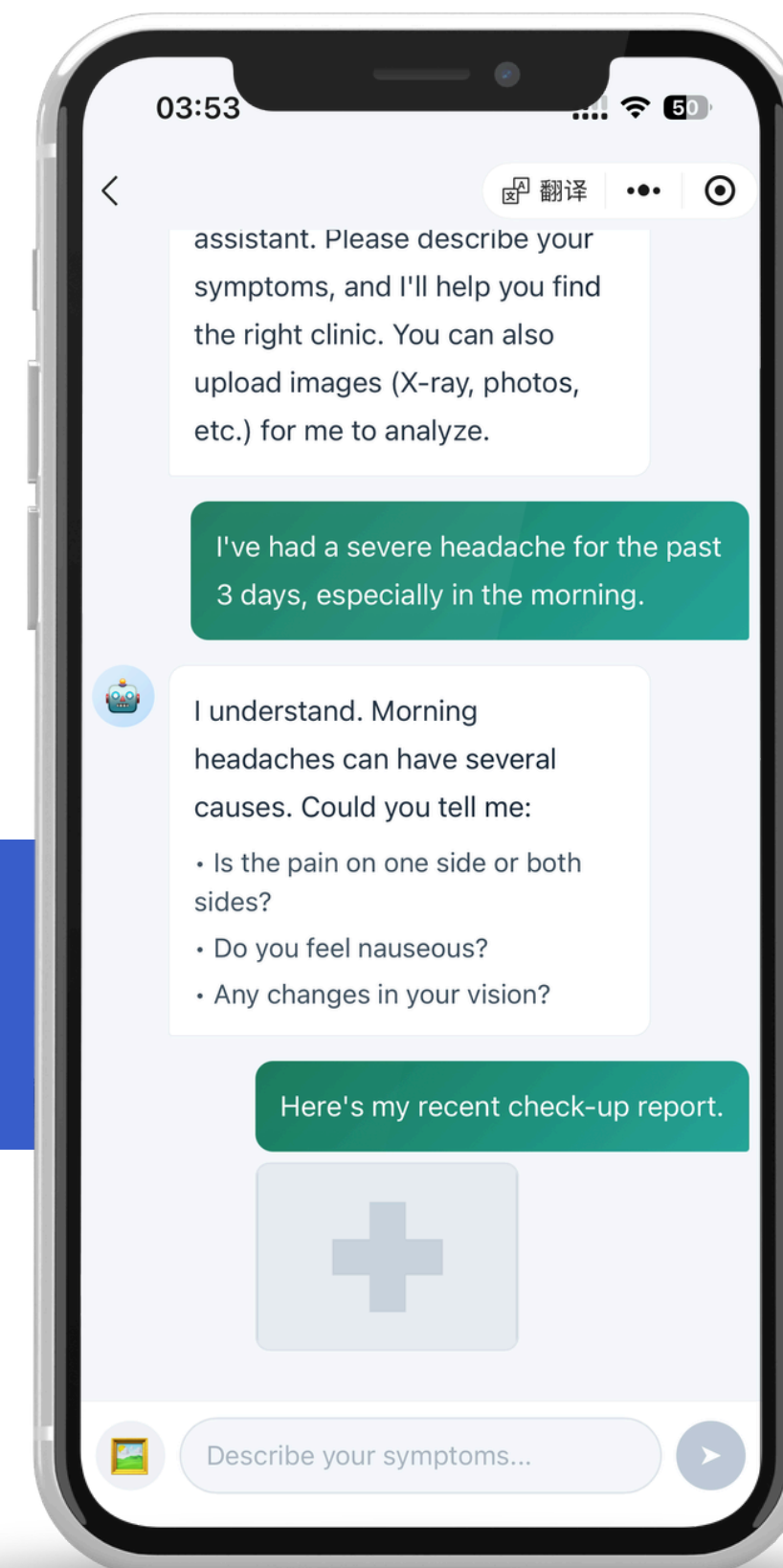


06 Final Prototype



Profile Page

AI Assistant





For final presentation

4/6/2026

07 Future Work

Technical aspects:

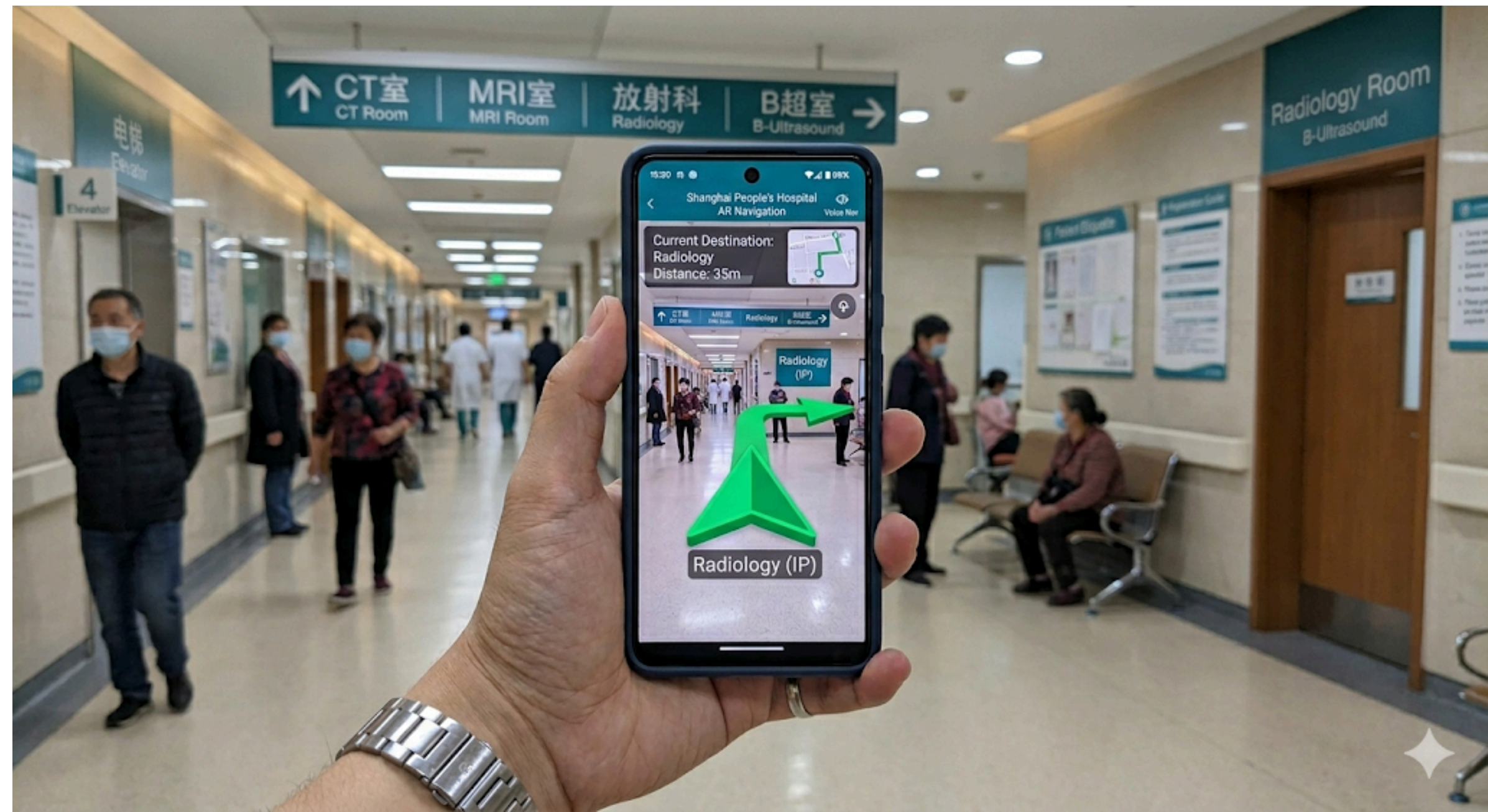
Check whether it can be integrated into a live system

Set up a mock server (simulated gateway): Write own backend simulator to emulate the APIs of major HIS vendors (e.g. open-source healthcare API standards or the HL7/FHIR protocols), and test this mini-programme's fault tolerance when faced with API delays or dirty data.

Features:

Integrates augmented reality technology to enhance the navigation experience.

Users can navigate the hospital in real time using their smartphone's camera.





For final presentation

4/6/2026

Thank You

